

Nationals Pitching Decision Support Tool

Product Requirements Document — v0.2

v0.2 update: Refined following initial Statcast prototype and Nationals bullpen diagnostic. Added team-level diagnosis, immediate-impact vs. development screening, acquisition context, and data-freshness requirements; deferred composite scoring until metric weighting can be justified.

1. Product Summary

The Nationals Pitching Decision Support Tool is a lightweight analytics product designed to help baseball operations staff diagnose bullpen weaknesses and identify pitchers whose measurable characteristics address those specific deficiencies.

Rather than beginning with a generic player ranking, the tool first compares the Nationals bullpen against MLB benchmarks, identifies areas of relative weakness, and then screens pitchers for characteristics that could address those gaps.

The initial prototype focuses on bullpen velocity, swing-and-miss ability, strikeout production, control, and contact suppression. It supplements Statcast performance data with roster, contract, transaction, and availability context to support deeper acquisition or development research.

Rather than simply ranking pitchers by ERA, strikeouts, or fastball velocity, the tool will combine pitch-level performance and physical characteristics to answer questions such as:

- Which pitchers generate unusually strong results for their velocity?
- Which pitchers possess high-velocity or high-movement pitches but have not yet produced strong traditional results?
- Which pitchers may have characteristics that could translate well into a bullpen role?
- Which available or potentially attainable pitchers resemble successful MLB relievers?

The product is intended as a **decision-support system**, not a replacement for scouts, coaches, biomechanical analysis, or front-office judgment.

2. Problem

A baseball organization must evaluate hundreds or thousands of pitchers across MLB, the minor leagues, free agency, waivers, and other organizations.

Traditional statistics such as ERA can obscure important information because they are affected by:

- defense,
- ballpark,
- sequencing,
- small sample size,
- role,
- competition level,
- and luck.

Pitch-level data can reveal characteristics that may be more predictive of future performance, but manually evaluating this information across a large player population is difficult.

The Washington Nationals' bullpen construction challenge provides a useful real-world use case.

One potential question is:

Can we identify pitchers who could add velocity, swing-and-miss capability, or other high-value pitch characteristics to the bullpen without relying solely on established high-cost relievers?

Early prototype analysis showed why diagnosis should precede player screening. In April–August 2026 public Statcast data, Washington ranked 25th in primary-fastball velocity, 26th in fastball whiff rate, and 30th in overall whiff rate . This suggests the acquisition problem is broader than simply adding velocity: the tool should identify pitchers who improve the bullpen's ability to generate swings and misses while considering control and contact quality.

3. Target User

Primary user

Baseball Operations Analyst / Quantitative Analyst

Secondary users

- Assistant General Manager
- Director of Pitching
- Pro Scouting Department
- Player Development Staff
- Pitching Coaches

4. Primary User Story

As a baseball operations analyst, I want to diagnose bullpen weaknesses and screen pitchers using pitch-level performance characteristics and roster context so that I can quickly identify players worthy of deeper scouting or acquisition analysis.

Example:

Show me relievers whose velocity, swing-and-miss ability, control, or contact suppression could address identified weaknesses in the Nationals bullpen.

5. MVP Question

The first version of the product should answer one narrow question well:

Which MLB pitchers appear to have characteristics that could make them attractive bullpen acquisition or development targets?

The MVP will initially focus on measurable characteristics available through public Statcast data.

6. MVP Inputs

Potential variables include:

Pitch characteristics

- Pitch type
- Average velocity
- Maximum velocity
- Spin rate
- Horizontal movement
- Vertical movement
- Release point
- Extension

Batter outcomes

- Whiff %
- Swing %
- Chase %
- Called-strike + whiff %
- Exit velocity allowed
- Hard-hit %
- Barrel %

Pitcher results

- Strikeout %
- Walk %
- ERA
- FIP or xERA where available
- Batters faced
- Innings pitched

Usage

- Pitch usage %
- Starter vs. reliever role
- Number of pitches thrown

Contextual Inputs

To support acquisition and roster evaluation, the MVP will supplement Statcast performance data with a lightweight contextual data layer that may include:

- Age
- Current organization
- 40-man roster status
- Options remaining, where publicly verifiable
- MLB service time
- Contract / control status
- Projected free-agent year
- Recent transaction history
- Current injury or availability notes

These fields are intended to provide decision context rather than estimate trade value. The MVP will not attempt to calculate acquisition cost or recommend specific transactions.

7. Initial Product Features

Feature 1 — Pitcher Filtering

Allow the user to filter the pitcher population using criteria such as:

- minimum pitches thrown,
- fastball velocity,
- whiff rate,
- strikeout rate,
- walk rate,
- pitch type,
- or bullpen role.

Example query:

Relievers averaging at least 96 mph with above-average whiff rates.

Screen Types

Immediate Impact Screen

Identifies pitchers already demonstrating strong velocity, swing-and-miss, strikeout production, and acceptable control.

Development / Upside Screen

Identifies pitchers with premium underlying traits but a potentially improvable weakness, such as elevated walk rate.

Feature 2 — Team Diagnostic

Compare the target team's bullpen against MLB bullpen benchmarks across metrics such as:

- primary-fastball velocity
- fastball whiff rate
- overall whiff rate
- strikeout rate
- walk rate
- contact quality

The diagnostic establishes which characteristics should drive the subsequent player screen.

A composite score may be explored in later versions after metric weights can be justified empirically. The MVP will favor transparent filters and comparisons over an arbitrary black-box score.

Feature 3 — Player Comparison

Allow the user to compare an individual pitcher against:

- MLB bullpen average
- Nationals bullpen baseline
- another pitcher
- selected candidate cohort

Example:

Metric	Pitcher	MLB Reliever Average
Fastball Velocity	97.2 mph	94.8 mph
Whiff %	31%	25%
K %	29%	23%
BB %	10%	8%

Feature 4 — Candidate Shortlist

Generate a small list of pitchers who warrant deeper scouting or acquisition analysis based on both performance fit and contextual information.

The shortlist should include:

- Pitcher
- Current Organization
- Average Fastball Velocity
- Fastball Whiff %
- Overall Whiff %
- Strikeout %
- Walk %
- Hard-Hit %
- Screen Type
- Contract / Control Context
- Availability / Injury Context
- Why He Fits WSH

Example output:

Pitcher	Organization	Avg FB Velo	Overall Whiff %	K %	BB %	Screen Type	Why He Fits WSH
Pitcher A	Team X	98.4	31%	34%	7%	Immediate Impact	Adds velocity, swing-and-miss, and control relative to the WSH bullpen baseline
Pitcher B	Team Y	96.8	30%	29%	12%	Development / Upside	Premium bat-missing traits with a control weakness that may be improvable
Pitcher C	Team Z	100.1	28%	30%	10%	Development / Upside	Extreme velocity directly addresses a team deficiency while offering additional strikeout upside

Shortlist inclusion does not mean the tool recommends acquiring the player. It indicates that the player's performance profile and contextual fit justify deeper scouting, roster, medical, and acquisition analysis.

The MVP should favor explainable comparisons and screening logic over a single composite score so that analysts can understand why each pitcher surfaced.

8. User Workflow

1. Analyst selects the team and analysis period.
 2. System establishes the bullpen population.
 3. System compares bullpen performance against MLB benchmarks.
 4. Analyst identifies one or more areas of need.
 5. Analyst chooses an Immediate Impact or Development/Upside screen.
 6. System filters qualifying pitchers using transparent thresholds.
 7. Analyst compares candidates against the team baseline.
 8. System displays roster/contract/availability context.
 9. Analyst creates a shortlist for deeper scouting or acquisition analysis.
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9. Data Source

The MVP will primarily use publicly available MLB Statcast data accessed through tools such as:

- Baseball Savant
- `pybaseball`
- MLB public data sources where appropriate

Additional public datasets may be added if needed.

The MVP will intentionally avoid proprietary scouting or biomechanical data.

10. Data Design & Freshness Requirements

Pitcher performance should be aggregated using a unique pitcher identifier rather than current organization so that trades do not split a player's analytical record.

Current organization, roster status, contract/control status, transaction history, and injury/availability status should be maintained as a separate contextual data layer because these attributes may change independently of the underlying performance sample.

The MVP should distinguish between:

- **Performance data** — derived from Statcast over the selected analysis period
- **Current context data** — current team, roster, contract, transaction, and availability information

Where contextual information cannot be reliably verified, the product should display the field as unknown or not verified rather than infer a value.

11. MVP Technology

Initial implementation:

- Python
- Jupyter Notebook or Google Colab
- pandas

- pybaseball
- matplotlib or Plotly
- Streamlit for a simple interactive interface
- Google Sheets — lightweight roster, contract, transaction, and availability context

The first version should prioritize analytical clarity over production-grade engineering.

12. Success Criteria

The MVP is successful if a user can:

1. Load and process MLB Statcast pitching data.
2. Distinguish starting and relief appearances.
3. Diagnose how a target bullpen differs from MLB benchmarks.
4. Screen pitchers using transparent performance criteria.
5. Separate immediate-impact and development/upside profiles.
6. Compare candidate metrics against the target bullpen baseline.
7. Add roster, contract, transaction, and availability context.
8. Produce a small shortlist and understand exactly why each pitcher surfaced.

A successful prototype should make it possible to explain:

“Here are several pitchers the model surfaced, here is why it surfaced them, and here is what a baseball operations analyst should investigate next.”

13. Non-Goals for Version 1

The MVP will **not** attempt to:

- predict exact future ERA,
- determine trade value,
- estimate contract value,
- replace professional scouting,
- analyze biomechanics,
- evaluate injury risk,
- recommend specific trades,
- model pitch design changes,
- or build a production MLB analytics platform.

These could become future iterations.

14. Future Versions

Potential extensions include:

V2 — Pitch Arsenal Optimization

Identify pitchers whose pitch mix may be suboptimal.

V3 — Pitcher Similarity

Find pitchers with similar pitch characteristics to successful MLB relievers.

V4 — Minor-League Identification

Extend the model to minor-league pitchers where data availability permits.

V5 — Acquisition Intelligence

Combine performance with:

- estimated acquisition cost
- trade-value modeling
- organizational depth and roster surplus
- comparable historical transactions
- salary and arbitration projections
- configurable team-specific acquisition constraints

15. Product Hypothesis

Pitch-level characteristics can reveal pitchers whose underlying abilities are stronger than their traditional statistics indicate.

A simple analytical tool combining velocity, movement, swing-and-miss performance, and contact quality could help baseball operations staff narrow a very large pitcher population into a manageable group for deeper evaluation.